Airand — Software Requirements Specification (SRS)

Oct 6, 2026 · @Francis · Phase 6 — Solution Requirements · v1.0 draft

1. Document control

Item	Detail
Document	Software Requirements Specification — Airand delivery platform
Version	1.0 draft
Status	For review by sponsor and development lead
Derived from	BRD v1.0 draft (6 Oct 2026); Future-State Design v0.1 (6 Oct 2026)
Structure	IEEE 830-style, lightened: introduction, overview, functional requirements by component, non-functional, data, interfaces, user stories, traceability

Version	Date	Change
1.0 draft	6 Oct 2026	First full draft: 50 functional requirements, 14 non-functional, 9 entities, 3 interfaces, 36 user stories in 7 epics, RTM · **1.1 draft, 6 Oct 2026:** sponsor corrections — fee paid to Airand at request and held until completion, with cyclist earnings credited only on completion (FR-06/07); tiered refund with 20% cancellation fee after assignment (FR-10); vendor delay detected by the system, not reported by the cyclist (FR-22/23, US-17); map and list views on the admin console (FR-51, US-37); open items OI-01 to OI-08 decided. **1.2 draft, 6 Oct 2026:** fee held by Airand in its own ledger, cyclist credited on completion (no third-party escrow); platform broadened from food to packages, with the pilot segment remaining food vendors; package limits added (FR-52, FR-53, US-38). **1.3 draft, 6 Oct 2026:** the dispatcher role renamed **system admin** (person: Chinedu) and merged with the former administrator class; the dispatcher board renamed **admin console**. Receiver page limited to two milestones and no cyclist appearance details (FR-38, US-25).

Identifier conventions: FR = functional requirement · NFR = non-functional requirement · DR = data requirement · IR = interface requirement · US = user story · BR / RUL / OBJ as in the BRD. Every FR names the BR and RUL it implements. Priorities are MoSCoW for the pilot release (R1); R2 marks the first post-pilot release.

2. Introduction

Purpose. This SRS specifies what the Airand platform must do to satisfy the 22 business requirements and 23 business rules in the BRD, as designed in the Future-State Design. It is written for the development team, the sponsor and the testers; the BRD remains the authority on *why*.

System scope. One core service and four front ends supporting fixed-rate, prepaid, system-assigned bicycle delivery in one zone: a vendor app, a cyclist app, a admin console and a receiver status page, integrated with a payment provider, a messaging gateway and a geocoding service. Out of scope for R1: consumer ordering, menus, live maps for receivers, route optimisation, batching, photo proof, ratings (BRD section 11).

User classes

Class	Who	Device and context	Technical comfort
Vendor	Small vendor or their staff sending packages (pilot segment: food vendors)	Android phone, often mid-cooking, poor data	Uses WhatsApp; little else
Cyclist	Airand cyclist on shift	Low-cost Android, outdoors, one hand, at night, intermittent data	Basic; needs large controls
System admin	Chinedu, Airand's operations admin; monitors the zone, handles exceptions and holds configuration rights	Laptop browser, indoors, full shift	Competent
Receiver	Vendor's customer	Any phone browser via a link; no account	Varies; must need zero explanation
Founder	Okoro Assumpta, sponsor: reports and pricing approvals only	Laptop browser	Competent
System	Scheduled and event-driven processes	—	—

Conventions. "Shall" = mandatory for the release stated. Timers and thresholds (60 s, 10 min, 2.5 km, ₦900) are configuration values with the pilot default given; the requirement is that the behaviour exists and the value is configurable without a code change.

3. System overview

The context diagram and component table are in the Future-State Design (section 7). The ownership rule that governs this SRS: **the core owns every rule and every record; front ends only present and capture.**

Component	Owns	Does not own
Core — Request & pricing	Zone boundary test, fixed-rate quote, payment gate, request expiry	Any UI
Core — Assignment engine	Cyclist availability, nearest-available selection, acknowledgement timeout, queue, reassignment	Navigation
Core — Lifecycle & record	Delivery states and events, code generation and validation, exception recording, immutability after Closed	Messaging transport
Core — Registry, statements, reporting	Vendors, cyclists, zones, hubs, rosters, wallets; weekly statements; KPI queries	Pay-out execution (bank)
Vendor app	Request form, wallet view, active and historic deliveries, statements	Pricing logic, zone logic
Cyclist app	Assignment acceptance, pickup and delivery confirmation, code entry, issue reporting, shift status, earnings view	Assignment decisions
Admin console	Live zone view, alerts, override and reassignment with reason, exception handling, daily close, reports	Rule changes (admin configuration only)
Receiver status page	Read-only status, ETA, cyclist first name, vendor phone	Any write

Request-to-delivery in system terms: vendor app POSTs a request → core prices and checks zone → payment confirmed into Airand's held-funds ledger → core activates and the assignment engine selects a cyclist → cyclist app receives the job and acknowledges → messaging gateway sends code and link to receiver → cyclist app posts pickup → cyclist app posts code → core validates, marks Delivered, recognises the fee as revenue and credits the cyclist's earning, writes events → statements and KPIs read from the same record.

4. Functional requirements — Core: request, pricing and payment gate

ID	Requirement	Traces to	Priority
FR-01	The system shall accept a delivery request from an authenticated vendor containing: drop-off address (free text), drop-off pin (lat/long, from geocoding or manual placement), receiver phone number, and an optional note of up to 140 characters. Pickup location shall default to the vendor's registered address.	BR-05; RUL-07	Must
FR-02	The system shall validate the receiver phone number as a Nigerian mobile number before accepting the request.	BR-05	Must
FR-03	The system shall geocode the drop-off address and test the resulting pin against the zone polygon. If outside, the system shall reject the request with reason "outside zone" and shall not take payment.	BR-08; RUL-07	Must
FR-04	The system shall reject a request submitted outside the zone's operating hours with reason "zone closed" and the next opening time.	RUL-08	Must
FR-05	The system shall quote the zone's fixed rate before payment and display it to the vendor; the rate shall be a zone-level configuration value, not derived from distance, time or order value.	BR-06; RUL-01	Must

ID	Requirement	Traces to	Priority
FR-06	The system shall create the request in state *Pending payment* and debit the fixed fee from the vendor's wallet into Airand's held-funds ledger; if the wallet balance is insufficient, the system shall offer a transfer-per-request reference. Held funds are not recognised as Airand revenue, and no cyclist earning is credited, until the delivery completes	BR-07; RUL-02, RUL-03	Must
FR-07	On confirmed payment (wallet debit or payment-provider callback), the system shall move the request to *Active* and trigger assignment immediately. When the delivery reaches *Delivered*, or *Failed* after pickup, the system shall move the held fee to recognised revenue and credit the assigned cyclist's per-delivery earning for that job; on *Expired* or *Cancelled* it shall refund from held funds per FR-10 and credit no cyclist earning	BR-07; RUL-02	Must
FR-08	If payment is not confirmed within the configured window (default 10 min), the system shall move the request to *Expired* and notify the vendor.	RUL-02	Must
FR-09	During a vendor's trial period (default 7 days from onboarding), the system shall price requests at ₦0 while still creating the request, payment record and full lifecycle.	BR-19; RUL-06	Should

ID	Requirement	Traces to	Priority
FR-10	The system shall allow a vendor to cancel a request before pickup. Before assignment the held fee shall be refunded in full to the wallet; after assignment but before pickup a configurable cancellation fee (default 20% of the rate, ₦180) shall be released to Airand and the balance refunded, because the cyclist is already riding to the vendor; after pickup the vendor cannot cancel and no refund is made. Every cancellation shall record who cancelled, the reason and the amounts released and refunded.	RUL-04, RUL-20	Must
FR-11	The system shall maintain a wallet per vendor with balance, top-up history and debit history, and shall accept top-ups by bank transfer confirmed via the payment provider.	RUL-03	Must
FR-12	The system shall allow a system admin to raise a request on a vendor's behalf (phone-in fallback), recorded as such, subject to the same validation and payment rules.	R-09 (BRD)	Should

Package scope (added 6 Oct 2026 when the platform was broadened from food to packages)

ID	Requirement	Traces to	Priority
FR-52	The system shall require the vendor to select a package category (food, documents, parcel, other) and confirm, by a checkbox on the request form, that the package fits the cyclist's bag, weighs no more than the configured limit (default 8 kg), has a declared value no more than the configured limit (default ₦50,000) and is not on the prohibited list; the category shall be stored on the delivery and shown to the cyclist.	BR-23; RUL-24	Must

ID	Requirement	Traces to	Priority
FR-53	The cyclist app shall let the cyclist refuse a package at pickup that visibly breaches the limits, with a reason; the request shall move to *Cancelled* with reason "package outside limits", the vendor refunded less the cancellation fee, and the system admin alerted.	BR-23; RUL-24	Must

5. Functional requirements — Core: assignment engine and delivery lifecycle

Assignment engine

ID	Requirement	Traces to	Priority
FR-13	The system shall hold a state for each cyclist from the set {Off shift, Available, Assigned, En route to pickup, Waiting at vendor, Delivering} and a last-known location with timestamp.	RUL-12	Must
FR-14	On a request becoming *Active*, the system shall select the cyclist in state *Available* in the same zone with the shortest straight-line distance from last-known location to the pickup; ties shall be broken by longest time since the cyclist's last assignment.	BR-09; RUL-11	Must
FR-15	The system shall move the request to *Assigned* only when the selected cyclist acknowledges in the cyclist app. If no acknowledgement is received within the configured timeout (default 60 s), the system shall mark the cyclist unresponsive for that request, move the request to *Reassigning*, exclude that cyclist, re-run selection and alert the system admin.	RUL-13	Must

ID	Requirement	Traces to	Priority
FR-16	If no cyclist is *Available*, the system shall move the request to *Awaiting cyclist*, show the vendor an expected wait, and alert the system admin after the configured threshold (default 3 min). When any cyclist becomes *Available*, the system shall re-run assignment for the oldest *Awaiting cyclist* request first.	RUL-14	Must
FR-17	The system shall allow a system admin to reassign any request to a named cyclist, to cancel a request, or to set a cyclist's state, each with a mandatory reason, and shall record the override as an event.	BR-10; RUL-15	Must
FR-18	The system shall exclude from selection any cyclist whose last-known location is older than a configured staleness limit (default 5 min) and shall flag them to the system admin.	RUL-11	Should

Delivery lifecycle and record

ID	Requirement	Traces to	Priority
FR-19	The system shall hold each delivery in exactly one of the states {Pending payment, Active, Awaiting cyclist, Assigned, Reassigning, Picked up, Delivered, Failed, Cancelled, Expired, Closed} and shall permit only the transitions defined in the Future-State Design lifecycle.	BR-11; RUL-21	Must
FR-20	The system shall write an immutable event for every state transition recording: delivery ID, from-state, to-state, timestamp (server), actor (vendor, cyclist, system admin or system, with ID) and reason where applicable.	BR-11, BR-22; RUL-21	Must
FR-21	On *Assigned*, the system shall generate a 4-digit one-time verification code unique among open deliveries in the zone, store it against the delivery, and send it to the receiver with a tracking link via the messaging gateway.	BR-12; RUL-16	Must

ID	Requirement	Traces to	Priority
FR-22	The system shall record *Picked up* when the assigned cyclist confirms handover, storing the timestamp; the cyclist's pickup wait shall be computed as Assigned → Arrived at vendor, and the vendor's handover time as Arrived → Picked up.	BR-11; RUL-17	Must
FR-23	The system shall record *Arrived at vendor* when the cyclist taps Arrived or when the cyclist's location is within 50 m of the pickup for 30 s, whichever is first. If *Picked up* is not recorded within the configured grace period (default 5 min) of arrival, the system shall automatically flag *Vendor delay*, exclude the excess from the cyclist's metrics, attribute it to the vendor's record, and alert the system admin if it persists beyond the configured limit (default 15 min). No cyclist action is required.	RUL-17; EX-05	Must
FR-24	The system shall record *Delivered* only when the code entered by the assigned cyclist matches the delivery's code; three consecutive mismatches shall lock code entry for that delivery and alert the system admin.	BR-12; RUL-18	Must
FR-25	The system shall allow the cyclist to record *Delivered without code* with a reason from a fixed list and the receiver's confirmed phone number; such deliveries shall be queued for system admin review before *Closed*.	RUL-18	Must
FR-26	The system shall allow the cyclist or system admin to record *Failed* with a reason from a fixed list (receiver unreachable, receiver refused, wrong address, cyclist incident, other) and a free-text note; the system shall notify the vendor and instruct return of the package.	BR-13; RUL-19; EX-06	Must

ID	Requirement	Traces to	Priority
FR-27	The system shall move *Delivered* deliveries to *Closed* automatically 24 hours after delivery if no review is pending, and *Failed* and *Delivered without code* deliveries to *Closed* on system admin confirmation. After *Closed* no field other than dispute notes shall be editable.	RUL-21	Must
FR-28	The system shall raise system admin alerts for: not acknowledged in 60 s; awaiting cyclist 3 min; not picked up 10 min after Assigned; not delivered 30 min after Picked up; vendor delay 15 min; code locked; stale cyclist location. Each alert shall name the delivery and the elapsed time.	RUL-23	Must
FR-29	The system shall retain all delivery, event, payment and exception records for at least 12 months and make them exportable by a system admin.	BR-22	Must

6. Functional requirements — Vendor app and Receiver status page

Vendor app

ID	Requirement	Traces to	Priority
FR-30	The vendor app shall let a vendor raise a request in a single screen: address field with geocoded suggestions and a map pin to adjust, receiver phone, optional note, the fixed fee displayed, and one confirm action that pays from the wallet.	BR-05, BR-06; FR-01, FR-05	Must
FR-31	The vendor app shall let a vendor save up to 20 frequent drop-off addresses and reuse them in one tap.	BR-05	Should

ID	Requirement	Traces to	Priority
FR-32	The vendor app shall show active deliveries with current state, cyclist first name, and elapsed time since request, updating without manual refresh.	BR-15	Must
FR-33	The vendor app shall show a history of completed, failed and cancelled deliveries filterable by date, with timestamps, fee, outcome and reason.	BR-15	Must
FR-34	The vendor app shall show the wallet balance, allow a top-up by displaying transfer instructions and a reference, and reflect confirmed top-ups within one minute of provider confirmation.	FR-11	Must
FR-35	The vendor app shall present the weekly statement (deliveries, fees, exceptions, opening and closing balance) and allow it to be shared as a PDF or image via WhatsApp.	BR-16; RUL-22	Must
FR-36	The vendor app shall allow cancellation of a request before pickup, showing whether a refund applies before the vendor confirms.	FR-10	Must
FR-37	The vendor app shall support onboarding by a system admin: vendor name, pickup address pinned on a map, phone, contact name, trial start date.	BR-18	Should

Receiver status page

ID	Requirement	Traces to	Priority
FR-38	The status page shall open from the unique link in the receiver's message without login and shall show: vendor name, delivery state in plain words, cyclist first name once assigned, estimated arrival once picked up, and the vendor's phone number. Progress shown to the receiver shall be limited to two milestones, Picked up and Delivered; payment, assignment and internal states shall not be shown.	BR-14; RUL-16	Must
FR-39	The status page shall display the receiver's verification code on the page as well as in the message, so a receiver who lost the message can still complete handover.	BR-12	Should
FR-40	The status page shall update its state without a reload at least every 30 seconds while the delivery is open, and shall show a terminal message once Delivered, Failed or Cancelled. The link shall stop revealing data 24 hours after the terminal state.	BR-14; NFR security	Must

7. Functional requirements — Cyclist app and Admin console

Cyclist app

ID	Requirement	Traces to	Priority
FR-41	The cyclist app shall let a cyclist start and end a shift; starting a shift sets the state to *Available* and begins location reporting at least every 60 seconds while on shift.	FR-13	Must
FR-42	The cyclist app shall present a new assignment as a full-screen prompt with vendor name, pickup distance and an Accept action, with an audible and vibrating alert, and shall show the seconds remaining to acknowledge.	FR-15	Must

ID	Requirement	Traces to	Priority
FR-43	The cyclist app shall show the current job in one screen: vendor address with a navigate button that opens the phone's maps app, receiver address and phone, note, and the single next action (Arrived, Picked up, Enter code, Report problem); arrival may also be detected automatically by geofence (FR-23).	BR-11; FR-22–26	Must
FR-44	The cyclist app shall queue actions taken while offline and submit them with their original device timestamps when connectivity returns, flagging the delay to the system admin if more than 2 minutes.	CON-03	Must
FR-45	The cyclist app shall show today's completed deliveries and an earnings estimate based on the retainer and bonus rules.	BR-04; RUL-10	Should

Admin console

ID	Requirement	Traces to	Priority
FR-46	The admin console shall show, for the selected zone, a live view of all open deliveries grouped by state, each with elapsed time in state, vendor, cyclist and alert status; and a panel of all cyclists with state, last location age and current job.	BR-10	Must
FR-47	The admin console shall surface alerts (FR-28) in a dedicated panel, each with one-click actions appropriate to it: reassign, call vendor, call receiver, call cyclist, cancel, mark failed, resolve.	BR-10, BR-13	Must
FR-48	The admin console shall provide a daily close view listing every delivery not yet terminal and every item pending review, and shall prevent closing the day while any remain.	FR-27	Should

ID	Requirement	Traces to	Priority
FR-49	The admin console shall provide reports for a selectable date range: deliveries by day and hour, request-to-assignment, pickup wait, delivery time (median and 90th percentile), failure and cancellation rates with reasons, vendor delays, deliveries per cyclist, cyclist utilisation, revenue and held-funds releases.	BR-17	Should
FR-50	The admin console shall provide system admin configuration for: zone polygon and hub, operating hours, fixed rate, cancellation fee, timers and thresholds, trial length, cyclist roster, and user accounts; every change shall be logged.	BR-01, BR-03; conventions	Must
FR-51	The admin console shall offer two live views of the same data, switchable in one click and remembered per user: a **list view** grouped by delivery state with distances, and a **map view** plotting cyclists, vendors with open requests and drop-offs on the zone polygon. Alerts shall surface identically in both.	BR-10	Must

8. Non-functional requirements

ID	Category	Requirement	Traces to	Priority
NFR-01	Performance	A vendor shall be able to complete a request (open app to payment confirmed from wallet) in under 30 seconds on a mid-range Android phone over a 3G connection.	OBJ-01; R-09	Must

ID	Category	Requirement	Traces to	Priority
NFR-02	Performance	Assignment selection shall complete within 5 seconds of a request becoming Active; the cyclist prompt shall appear within 10 seconds.	OBJ-01	Must
NFR-03	Performance	The receiver's code and link message shall be dispatched to the gateway within 10 seconds of Assigned.	BR-12	Must
NFR-04	Availability	Core services shall be available 99.5% of the time during zone operating hours, measured monthly; planned maintenance shall fall outside operating hours.	OBJ-01	Must
NFR-05	Resilience	The cyclist app shall remain usable for the current job with no connectivity: the job details, navigation hand-off and action buttons shall work offline and sync later (FR-44).	CON-03	Must
NFR-06	Device support	The cyclist and vendor apps shall run on Android 10 or later with 2 GB RAM and a 5-inch screen; all primary actions shall be reachable with one thumb and have touch targets of at least 48 dp.	CON-03	Must
NFR-07	Usability	A new cyclist shall complete a training delivery end to end after no more than 15 minutes of instruction; a vendor shall raise a first request after a 5-minute walkthrough.	S-01, S-03 needs	Should

ID	Category	Requirement	Traces to	Priority
NFR-08	Security	Receiver phone numbers shall be visible only to the assigned cyclist (for the duration of the job), the system admin and the vendor who supplied them; they shall be masked in reports and exports.	BR-22; data protection	Must
NFR-09	Security	All traffic shall be encrypted in transit; wallet operations and configuration changes shall require authenticated users with role-based access (vendor, cyclist, system admin).	CON-04	Must
NFR-10	Security	The receiver status link shall be unguessable (at least 128 bits of entropy) and shall expire 24 hours after the delivery reaches a terminal state.	FR-40	Must
NFR-11	Auditability	Every state transition, override, configuration change and payment event shall be recorded with actor and server timestamp and shall be immutable.	BR-22; FR-20	Must
NFR-12	Data retention	Delivery, event, payment and exception records shall be retained for at least 12 months and exportable in CSV for regulatory inspection and dispute handling.	BR-22; CLRD	Must
NFR-13	Scalability	The design shall support at least 5 zones, 50 cyclists and 1,000 deliveries a day without architectural change, so that a second zone is a configuration task.	BRD scope (deferred zones)	Should

ID	Category	Requirement	Traces to	Priority
NFR-14	Localisation	All user-facing text shall be in plain English with Nigerian conventions (₦, +234 numbers, 24-hour times); the vendor and cyclist apps shall avoid jargon ("Package picked up", not "Status: PICKED_UP").	S-01, S-03	Must

9. Data requirements

Nine entities carry the whole model; the Phase 8 ERD and data dictionary will detail attributes, types and keys. Cardinalities: a Zone has one Hub and many Vendors and Cyclists; a Vendor has one Wallet and many Deliveries; a Delivery has one Vendor, at most one Cyclist at a time (history via Events), one Payment, many Events; a Statement groups a Vendor's Deliveries for one week.

ID	Entity	Key attributes the system must hold	Why
DR-01	Zone	ID, name, polygon, hub location, operating hours, fixed rate, timer configuration	FR-03, FR-04, FR-05, FR-50
DR-02	Vendor	ID, name, pickup address and pin, phone, contact, trial start, status, zone	FR-01, FR-09, FR-37
DR-03	Wallet	Vendor ID, balance, ledger of top-ups and debits with provider references	FR-06, FR-11, FR-34
DR-04	Cyclist	ID, name, phone, zone, state, last location + timestamp, shift status, roster, pay terms	FR-13, FR-14, FR-41, FR-45
DR-05	Delivery	ID, vendor, zone, drop-off address + pin, receiver phone (masked in exports), note, state, assigned cyclist, code (hashed), fee, created/activated/assigned/picked/delivered timestamps, outcome and reason	FR-01, FR-19–27
DR-06	Event	Delivery ID, from-state, to-state, actor type and ID, server timestamp, device timestamp, reason, free text	FR-20, NFR-11

ID	Entity	Key attributes the system must hold	Why
DR-07	Payment	Delivery ID, vendor, amount, method (wallet/transfer), provider reference, fee status (held / released / refunded / partly refunded), amount released, amount refunded, cancellation fee applied, held and settled timestamps	FR-06–08, FR-10
DR-08	Statement	Vendor ID, week start/end, delivery count, fees total, exceptions, opening and closing balance, generated timestamp	FR-35; RUL-22
DR-09	Alert	Delivery or cyclist ID, type, raised timestamp, resolved timestamp, resolver, action taken	FR-28, FR-47

Derived metrics (computed from Delivery and Event, not stored as inputs): request-to-assignment, pickup wait, delivery time, failure and cancellation rates, deliveries per cyclist per shift, utilisation (time in Assigned/En route/Delivering over time on shift).

10. Interface requirements

ID	Interface	Direction	Requirement	Traces to
IR-01	Payment provider	Core → provider; provider → core (callback)	The core shall talk to the payment provider through a single adapter with three operations: top-up (bank transfer with reference), refund-to-bank (optional; wallet refunds are internal), and status. Holding and release are Airand's own ledger, not the provider's. For the pilot the adapter targets a fictional provider, "PayHold", whose contract is documented in the adapter; a real Nigerian provider offering transfer-reference confirmation is swapped in by implementing the same three operations, with no change to the core. The core shall accept a signed confirmation callback that moves the request to Active (FR-07) and shall poll status at least every 60 s if no callback arrives beforethe request expires.	FR-06–08, FR-11

ID	Interface	Direction	Requirement	Traces to
IR-02	Messaging gateway	Core → gateway	The core shall send templated messages via WhatsApp Business API with SMS fallback if WhatsApp delivery is not confirmed within 30 s: receiver code + link on Assigned; optional status messages to vendors. Delivery receipts shall be recorded against the Event. Message templates shall be configurable.	FR-21, NFR-03
IR-03	Maps and geocoding	Core → provider; apps → phone maps	The core shall geocode free-text addresses to coordinates and offer suggestions during entry (FR-30); the zone test shall run on coordinates. The cyclist app shall hand navigation to the phone's installed maps app via a deep link rather than embedding a map. Provider shall be swappable; straight-line distance for assignment needs no provider call.	FR-03, FR-30, FR-43

Not an interface in R1: marketplace APIs for inbound jobs (deferred), bank pay-out API for cyclist wages (manual transfers from a weekly report in R1).

11. User stories

36 stories in seven epics. Each has Gherkin acceptance criteria, a MoSCoW priority and a release (R1 = pilot). Stories are written to be pasted into Jira; the FR column is the link back to this document.

Epic E1 — Vendor requests a delivery

ID	Story	Acceptance criteria	FR	Priority	Release
US-01	As a vendor, I want to request a delivery by entering the address and the receiver's phone so that I don't have to phone around for a rider.	Given I am logged in and my wallet has ≥ ₦900, when I enter a valid in-zone address and a valid Nigerian number and confirm, then the request is created, ₦900 is debited from my wallet and held by Airandand I see "Finding your cyclist" within 5 s.	FR-01, 02, 05, 06, 07, 30	Must	R1
US-02	As a vendor, I want to see the delivery fee before I confirm.	Given I have entered a drop-off, when the address is in zone, then the fixed rate is shown before any confirm button is enabled; and the rate is identical at 13:00 and 21:00.	FR-05	Must	R1
US-03	As a vendor, I want to be told immediately if an address is outside the zone so that I can decline the order or correct the address.	Given I enter an address whose pin falls outside the polygon, when I leave the field, then I see "Outside our delivery zone" with the zone map and no payment is attempted.	FR-03	Must	R1
US-04	As a vendor, I want to reuse addresses I deliver to often so that repeat orders take one tap.	Given I have saved an address, when I start a request, then I can select it from a list and the pin and phone are pre-filled.	FR-31	Should	R1
US-05	As a vendor, I want to top up my wallet by bank transfer so that requests are paid instantly.	Given I open the wallet, when I choose Top up, then I see account details and a unique reference; and when the transfer is confirmed by the provider, my balance updates within 60 s.	FR-11, 34; IR-01	Must	R1
US-06	As a vendor, I want to pay per request by transfer when my wallet is empty so that I am never blocked.	Given my balance is below the fee, when I confirm a request, then I am shown a transfer reference; and when payment confirms within 10 min the request activates, otherwise it expires and I am told.	FR-06, 08	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-07	As a vendor, I want to cancel a request before pickup so that I don't pay for an order my customer cancelled.	Given a request is Active or Awaiting cyclist, when I cancel, then the full ₦900 returns to my wallet; given it is Assigned, when I cancel, then I am shown "₦180 cancellation fee, ₦720 refunded" before I confirm and the refund lands within 60 s; given it is Picked up, then I cannot cancel and am told to call the system admin.	FR-10, 36	Must	R1

Epic E2 — System assigns a cyclist

ID	Story	Acceptance criteria	FR	Priority	Release
US-09	As the system, I want to assign the nearest available cyclist automatically so that no one has to dispatch by hand.	Given a request becomes Active and two cyclists are Available at 300 m and 900 m, when assignment runs, then the 300 m cyclist is prompted within 5 s; given both are at 300 m, then the one idle longest is prompted.	FR-14	Must	R1
US-10	As the system, I want to move on if a cyclist doesn't respond so that one silent phone doesn't stall the order.	Given a cyclist is prompted, when 60 s pass without Accept, then the request goes to Reassigning, that cyclist is excluded, the next nearest is prompted and the system admin sees an alert.	FR-15	Must	R1
US-11	As a vendor, I want to know when no cyclist is free so that I can manage my customer's expectations.	Given no cyclist is Available, when my request activates, then I see "All cyclists busy — expected wait about N min" and the request is queued; when a cyclist frees up, then the oldest queued request is assigned first.	FR-16	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-12	As a system admin, I want to reassign a delivery to a specific cyclist with a reason so that I can fix what the rule gets wrong.	Given an open delivery, when I choose Reassign, pick a cyclist and enter a reason, then the previous cyclist is released, the new one is prompted and an override event is recorded with my ID and reason.	FR-17	Must	R1
US-13	As the system, I want to ignore cyclists whose location is stale so that I don't assign to a phone that has gone dark.	Given a cyclist's last location is older than 5 min, when assignment runs, then they are skipped and flagged on the admin console.	FR-18	Should	R1

Epic E1b — Package scope

ID	Story	Acceptance criteria	FR	Priority	Release
US-38	As a vendor, I want to state what I'm sending and confirm it fits the limits so that the cyclist can carry it and I'm not charged for a refused pickup.	Given I am on the request form, when I have not chosen a category or ticked the limits confirmation, then Confirm is disabled; when I choose "Documents" and tick, then the request proceeds and the cyclist sees "Documents" on the job. Given the cyclist refuses at pickup with reason "over limit", then the delivery is Cancelled, I am refunded ₦720 and shown the reason.	FR-52, FR-53	Must	R1

Epic E3 — Cyclist fulfils a delivery

ID	Story	Acceptance criteria	FR	Priority	Release
US-14	As a cyclist, I want a loud, clear prompt for a new job so that I never miss one while waiting at the hub.	Given I am Available and on shift, when I am selected, then my phone sounds and vibrates, shows vendor name and distance full-screen with a 60 s countdown, and Accept is a single large button.	FR-42	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-15	As a cyclist, I want to navigate to the vendor and then the receiver with one tap so that I don't fumble with addresses while riding.	Given I have a job, when I tap Navigate, then my phone's maps app opens with the current destination; after Picked up, the destination switches to the receiver.	FR-43; IR-03	Must	R1
US-16	As a cyclist, I want to confirm pickup when the vendor hands me the package so that the record shows exactly when I left.	Given I am at the vendor, when I tap Picked up, then the delivery becomes Picked up with a timestamp, the vendor sees it, and the receiver's ETA appears on their page.	FR-22	Must	R1
US-17	As the system, I want to detect when a vendor delays handover so that the wait is attributed to the vendor and never to the cyclist, without the cyclist having to do anything.	Given the cyclist's arrival is recorded (tap or 50 m geofence), when 5 min pass without Picked up, then a Vendor delay flag is written with the time, the cyclist's metrics stop at arrival, the vendor's handover time runs from arrival, and the system admin is alerted at 15 min.	FR-23	Must	R1
US-18	As a cyclist, I want to close the delivery by entering the receiver's code so that there is proof I delivered to the right person.	Given I am at the drop-off, when I enter the correct 4-digit code, then the delivery becomes Delivered and I return to Available; when I enter a wrong code three times, then entry locks and the system admin is alerted.	FR-24	Must	R1
US-19	As a cyclist, I want a way to complete a delivery when the receiver can't show the code so that a lost message doesn't strand a package.	Given the receiver is present but has no code, when I choose Delivered without code, confirm their phone number and pick a reason, then the delivery is marked for system admin review and I am released.	FR-25	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-20	As a cyclist, I want to report that I cannot reach the receiver so that the procedure runs and I'm not blamed.	Given I have waited 10 min and made two calls, when I tap Can't reach receiver, then the system admin is alerted to call the vendor's number; when the system admin confirms failure, then the delivery is Failed with reason and I am told to return the package.	FR-26	Must	R1
US-21	As a cyclist, I want the app to keep working when I lose signal so that I can still finish the job.	Given I am offline, when I tap Picked up or enter a code, then the action is stored with the device time and the screen confirms it; when I regain signal, then it syncs and, if more than 2 min late, the system admin sees a sync-delay flag.	FR-44; NFR-05	Must	R1
US-22	As a cyclist, I want to start and end my shift in the app so that I only get jobs when I'm working.	Given I tap Start shift, then I become Available and my location is shared every 60 s; given I tap End shift with no open job, then I become Off shift and location sharing stops.	FR-41	Must	R1
US-23	As a cyclist, I want to see today's deliveries and my earnings so that I trust my pay.	Given I open Today, then I see each completed delivery with time, and an earnings estimate from retainer plus bonus above the daily threshold.	FR-45	Should	R1

Epic E4 — Receiver follows the delivery

ID	Story	Acceptance criteria	FR	Priority	Release
US-24	As a receiver, I want a message with a code and a link as soon as a cyclist is assigned so that I know the order is coming and how to confirm it.	Given a delivery becomes Assigned, when 10 s pass, then a WhatsApp message with vendor name, 4-digit code and link has been sent; if WhatsApp is not confirmed in 30 s, then an SMS is sent instead.	FR-21; IR-02	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-25	As a receiver, I want to open the link and see where my package is so that I don't have to call the vendor.	Given I open the link, then without logging in I see the state in plain words, the cyclist's first name once assigned, an ETA once picked up, and the vendor's number; progress shows only two milestones, Picked up and At your door, with no payment or assignment steps; the page updates itself at least every 30 s.	FR-38, 40	Must	R1
US-26	As a receiver, I want my code shown on the page too so that I can still confirm if I deleted the message.	Given I open the link while the delivery is open, then the code is visible on the page; after the delivery closes, the code is no longer shown.	FR-39	Should	R1

Epic E5 — System admin monitors and handles exceptions

ID	Story	Acceptance criteria	FR	Priority	Release
US-27	As a system admin, I want one live view of every open delivery and every cyclist so that I can see the zone at a glance.	Given I open the board, then I see deliveries grouped by state with elapsed time, and cyclists with state, location age and current job, refreshing without reload.	FR-46	Must	R1
US-28	As a system admin, I want alerts with the right actions attached so that I handle exceptions in one click.	Given a delivery is not picked up 10 min after Assigned, then an alert appears naming it with Call cyclist, Call vendor and Reassign actions; when I act, then the alert records my action and resolution time.	FR-28, 47	Must	R1
US-29	As a system admin, I want to raise a request for a vendor who phones in so that a vendor without data can still get a cyclist.	Given a vendor calls, when I enter their request on their behalf, then it follows the same validation and payment rules and is marked "raised by system admin".	FR-12	Should	R1
US-30	As a system admin, I want to review deliveries completed without a code before they close so that I catch wrong-recipient cases.	Given a Delivered without code delivery exists, when I open the review queue, then I see cyclist, reason and phone confirmation and can Close or mark Failed with a note.	FR-25, 27	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-31	As a system admin, I want a daily close that won't let me finish with loose ends so that every delivery has an outcome.	Given it is end of shift, when I open Close day, then I see every non-terminal delivery and pending review; Close is disabled until the list is empty.	FR-48	Should	R1
US-37	As a system admin, I want to switch between a list and a map of the zone so that I can work by state when it's busy and by geography when I'm placing cyclists.	Given I am on the board, when I toggle Map, then I see the zone polygon, every on-shift cyclist, vendors with open requests and open drop-offs, with alerts highlighted; when I toggle List, then I see the state-grouped view; my choice persists at next login.	FR-51	Must	R1

Epic E6 — Vendor sees history and gets paid-for records

ID	Story	Acceptance criteria	FR	Priority	Release
US-32	As a vendor, I want to see my active deliveries and their status so that I can answer my customer without calling anyone.	Given I have open deliveries, when I open the app, then each shows its state, cyclist first name and elapsed time, updating live.	FR-32	Must	R1
US-33	As a vendor, I want a history of all my deliveries with outcomes so that disputes are settled from the record.	Given I open History, then I can filter by date and see each delivery's timestamps, fee, outcome and reason.	FR-33	Must	R1
US-34	As a vendor, I want a weekly statement I can share so that I know what delivery cost me.	Given Monday 06:00, then my statement for the previous week is available with deliveries, fees, exceptions and balances, and I can share it to WhatsApp as PDF.	FR-35	Must	R1

Epic E7 — Management configures and reports

ID	Story	Acceptance criteria	FR	Priority	Release
US-35	As a system admin, I want to configure the zone, rate, hours and timers so that changing a rule is not a code change.	Given I edit the fixed rate, when I save, then new requests use the new rate immediately, open requests keep theirs, and the change is logged with my ID.	FR-50	Must	R1

ID	Story	Acceptance criteria	FR	Priority	Release
US-36	As the founder, I want reports on volume, timings, failures and utilisation by hour so that I size the pool and judge the pilot.	Given I pick a date range, then I see deliveries by day and hour, median and 90th-percentile request-to-assignment, pickup wait and delivery time, failure and cancellation reasons, deliveries per cyclist and utilisation, and revenue.	FR-49	Should	R1

12. Requirements Traceability Matrix

Every business requirement is covered by at least one functional requirement and one story; every FR traces back to a BR. The Wireframe and UAT columns are filled in Phases 7 and 8. Gaps would show as empty cells; there are none at FR or story level.

Objective	BR	FR / NFR	User stories	Wireframe (Ph 7)	UAT case (Ph 8)
OBJ-01 Reliable at peak	BR-01 Zones	FR-03, FR-50; DR-01	US-03, US-35	—	—
OBJ-01	BR-02 Stationed pool at peak	FR-41; DR-04	US-22	—	—
OBJ-04 Viable volume	BR-03 Size pool to demand	FR-49, FR-50	US-35, US-36	—	—
OBJ-01	BR-04 Retainer pay	FR-45; DR-04	US-23	—	—
OBJ-01	BR-05 Vendor requests	FR-01, 02, 30, 31; NFR-01	US-01, US-04	—	—
OBJ-03 Predictable cost	BR-06 Fixed rate shown first	FR-05	US-02	—	—
OBJ-03	BR-07 Payment before activation	FR-06, 07, 08, 11, 34; IR-01	US-05, US-06	—	—
OBJ-01	BR-08 Decline outside zone	FR-03	US-03	—	—
OBJ-01	BR-09 Automatic assignment	FR-13, 14, 15, 16, 18; NFR-02	US-09, 10, 11, 13	—	—
OBJ-01	BR-10 System admin sees all, can override	FR-17, 46, 47, 51	US-12, 27, 28, 37	—	—

Objective	BR	FR / NFR	User stories	Wireframe (Ph 7)	UAT case (Ph 8)
OBJ-02, 05	BR-11 Pickup and delivery recorded	FR-19, 20, 22, 43	US-15, US-16	—	—
OBJ-05 Prove every delivery	BR-12 Receiver code	FR-21, 24, 39; NFR-03; IR-02	US-18, 24, 26	—	—
OBJ-05	BR-13 Exception procedure	FR-23, 25, 26, 28, 47	US-17, 19, 20, 28, 30	—	—
OBJ-02 Fast	BR-14 Receiver status	FR-38, 40; NFR-10	US-25	—	—
OBJ-05	BR-15 Vendor active and history	FR-32, 33	US-32, US-33	—	—
OBJ-05	BR-16 Weekly statement	FR-35; DR-08	US-34	—	—
OBJ-04	BR-17 Management reporting	FR-49	US-36	—	—
OBJ-04	BR-18 One-day onboarding	FR-37	(admin task; US-35)	—	—
OBJ-04	BR-19 Free trial	FR-09	US-08	—	—
OBJ-06 Legal and safe	BR-20 Trained, equipped cyclists	NFR-07	(operational; no system story)	—	—
OBJ-06	BR-21 CLRD licence, complaints	NFR-12; FR-29	(operational; records exportable)	—	—
OBJ-05, 06	BR-22 12-month records	FR-20, 29; NFR-08, 11, 12	(system-wide)	—	—
—	Cross-cutting (CON-03)	FR-44; NFR-05, 06, 14	US-21	—	—
—	Cross-cutting (R-09 fallback)	FR-12	US-29	—	—
—	Cross-cutting (cancellation RUL-04/20)	FR-10, 36	US-07	—	—
—	Cross-cutting (daily close RUL-21)	FR-27, 48	US-31	—	—

Coverage check: 23 BRs → 53 FRs + 14 NFRs → 38 stories. BR-23 (package limits) → FR-52, FR-53 → US-38. BR-20 and BR-21 are operational requirements with no user-facing story; they are satisfied by procedures and by records the system makes exportable, and will

appear in UAT as checklist items rather than test scripts.

Wireframe column (Phase 7, 6 Oct 2026). Screens are on the *Airand — Wireframes* canvas. Each Must story with a user-facing screen now maps to a wireframe; BR-20/21/22 remain procedural or system-wide.

Wireframe	Component	FRs shown	Stories	BRs
WF-01 Live board (list / map toggle)	Admin console	FR-15, 16, 18, 23, 28, 46, 47, 51	US-10, 11, 13, 17, 27, 28, 37	BR-09, 10, 13
WF-02 Review queue & close day	Admin console	FR-25, 26, 27, 48	US-20, 30, 31	BR-13, 22
WF-03 Request a delivery	Vendor app	FR-01–07, 30, 31, 52	US-01, 02, 03, 04, 38	BR-05, 06, 07, 08, 23
WF-04 Active deliveries & cancel	Vendor app	FR-10, 32, 36	US-07, 32	BR-15
WF-05 Wallet & weekly statement	Vendor app	FR-11, 33, 34, 35	US-05, 33, 34	BR-07, 15, 16
WF-06 New job prompt	Cyclist app	FR-14, 15, 42	US-09, 14	BR-09
WF-07 Current job	Cyclist app	FR-22, 23, 24, 25, 26, 43, 53	US-15, 16, 18, 19, 20	BR-11, 12, 13, 23
WF-08 Today & shift	Cyclist app	FR-41, 45	US-22, 23	BR-02, 04
WF-09 Status page	Receiver page	FR-21, 38, 39, 40	US-24, 25, 26	BR-12, 14

Not drawn in R1 wireframes: system admin configuration (FR-50, US-35), reports (FR-49, US-36; Phase 8 BI dashboard), system admin-raised request (FR-12, US-29), offline sync indicator (FR-44, US-21).

UAT column (Phase 8b). The tests that prove each business requirement are in *Airand — UAT Pack*, section "Traceability": 31 scenarios for the 30 Must stories, 12 exception and boundary tests, and 6 operational checks. 19 of 23 BRs are proved before go-live; BR-04, 17, 18 and 19 rest on Should stories scheduled after the pilot.

13. Open items and decisions

Item	Question	Decision	Owner	Phase
OI-01	Map or state-grouped list on the admin console?	**Both**, as one-click toggle views of the same data (FR-51, US-37). List for busy periods, map for positioning cyclists.	BA + founder	Closed
OI-02	Native Android or web app?	**Vendor app: progressive web app** (installable from a link, no store listing, updates instantly, cheap to build; offline needs are light). **Cyclist app: native Android**, because background location every 60 s, loud full-screen prompts and a reliable offline action queue need native capabilities. Admin console and receiver page: web.	BA (recommendation)	Closed
OI-03	Which payment provider?	**Fictional provider "PayHold"** behind a five-operation adapter (top-up, hold, release, refund, status). The pilot design and tests run against PayHold's documented contract; implementation swaps in a real Nigerian provider by writing one adapter, with no change to the core (IR-01).	BA	Closed

Item	Question	Decision	Owner	Phase
OI-04	WhatsApp Business API lead time and cost?	Plan **4–6 weeks** before launch: Meta business verification typically 1–3 weeks; onboarding via a business solution provider ~1 week; message template approval usually within 24–48 h but allow a week for rejections and resubmission. Budget ≈ ₦15–₦25 per utility message and ≈ ₦5 per SMS fallback [Assumption]; the economics model's ₦10/delivery is kept as the blended estimate. Start verification at project kickoff.	Founder	Closed
OI-05	Final attributes, types and keys for the nine entities?	Phase 8 ERD and data dictionary. Decided now: UUID primary keys; all timestamps stored in UTC with the device timestamp kept separately on Events; Events append-only; receiver phone encrypted at rest; codes stored hashed.	BA	Phase 8

Item	Question	Decision	Owner	Phase
OI-06	KPI definitions and synthetic data?	Phase 8: each KPI defined once as SQL against the data model; synthetic dataset of 90 days, 8 vendors (2 anchors), 10 cyclists, demand ramping 30 → 70 deliveries/day with a 7–11 PM peak, 4% failures, 3% cancellations, clearly labelled [Simulated].	BA	Phase 8
OI-07	UAT coverage?	Phase 8: one scenario per Must story (28), checklist items for BR-20 and BR-21, one negative test per exception EX-01 to EX-06; RTM wireframe and UAT columns completed.	BA	Phase 8
OI-08	Are the timers right?	Pilot defaults stand (60 s acknowledge, 3 min queue alert, 5 min handover grace, 10 min pickup alert, 15 min delay alert, 30 min delivery alert, 10 min payment expiry); all configurable (FR-50). Review after 4 weeks of pilot data.	Founder	Post-pilot